

NUCLEAR REBUILD

Complete Project Explanation in Simple Terms (ML, NLP & Modalities)

1. Executive Summary

This project is an AI-powered verification system designed to detect fake news and calculate a **0–100 Trust Score** for news articles related to nuclear power and public safety. It allows users to verify information from **3 input formats**: raw text, web links (URLs), or news screenshots/images.

2. Understanding ML & NLP in Simple Terms

What is NLP (Natural Language Processing)?

NLP is the branch of Artificial Intelligence that teaches computers how to read, understand, and analyze human languages (words, sentences, and tone). In this project, NLP works like a digital editor that reads news text, cleans up clutter, detects sensational or emotional words, and extracts key facts.

What is ML (Machine Learning)?

ML is the technology that allows computers to learn patterns from data instead of following rigid rules. We trained our ML model on thousands of real and fake news articles. By looking at these examples, the ML model learned the subtle patterns of legitimate journalism versus deceptive misinformation.

3. How the 3 Input Modalities Work

■ 1. Text Input (Direct Articles)

The user pastes news text directly into the application. The system cleans the text (removing extra spacing, special symbols, and formatting) and passes it directly to the NLP models for fake news detection.

■ 2. Link / URL Input (Web Articles)

The user provides a website link (e.g., Reuters, BBC, or an unverified blog). The system automatically fetches and extracts the full article text while also assessing the **reputation of the domain** (e.g., trusted news agency vs. unknown site).

■ ■ 3. Image Input (Screenshots & Social Media Posts)

The user uploads a screenshot of a news headline or social media graphic. An **OCR (Optical Character Recognition)** model reads the pixels in the image and converts the text into digital words, which are then passed to the NLP model.

4. All Models Used in the System (Simplified)

Model Name	Type	Role in Simple Terms
RoBERTa / DeBERTa	Deep Learning (Transformer NLP)	Main Classifier: Reads text context & classifies it as Real or Fake based on learned news patterns.
EasyOCR	Computer Vision + OCR Model	Image Reader: Extracts printed/written text from uploaded news screenshots.

VADER & TextBlob	Rule-based NLP Sentiment Models	Tone Analyzer: Measures whether the text is calm/neutral (credible) or overly emotional/dramatic (suspicious).
Domain Credibility Scorer	Heuristic Rule Engine	Source Check: Assigns trust scores based on official domain reputation (e.g. government, news wire vs. clickbait blogs).
Multi-Factor Trust Engine	Ensemble Scoring Engine	Score Calculator: Combines 4 signals into a final 0–100 Trust Score .

5. How the 0–100 Trust Score is Computed

The system does not rely on a single score. It calculates 4 individual weights into a final score:

- **45% ML Model Confidence:** Output probability from the RoBERTa classifier.
- **25% Source Credibility:** Reputation ranking of the news source/domain.
- **20% Fact Corroboration:** Match against verified official nuclear databases.
- **10% Tone Neutrality:** Absence of clickbait / sensational emotional language.